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## (54) Title: **A system for swarm of drone for crowed behaviour analysis**

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(57) Abstract:

A system 100 to analyze crowed behaviors using swarm of drones 102 includes a network of sensors 104 including a stereo camera 104-A, LiDAR 104-B, inertial measurement unit 104-C, and a GPS module 104-D; a human machine interface 150; and an onboard controller 108 executing simultaneous localization and mapping, including at least one database, one or more processors to receive commands from the human-machine interface 150 to take-off and position the drone 102 at one or more target locations; maintain peer-to-peer links within the swarm of drone network; receive and record data from sensors 104 for processing real-time images for preliminary crowed flow vectors; transmit processed data to centralized decision module 110 and collaborative SLAM module fusion 112 using communication transceiver 120; detection and validation of anomalies to confirm cored hazards and emergency response; and continue mapping, avoidance of peer-to-peer collision, and failover handling if centralized decision module 110 link is lost.

**FORM 2**

THE PATENTS ACT, 1970

(39 of 1970)

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The Patent Rules, 2003

**COMPLETE SPECIFICATION**

(See sections 10 & rule 13)

**1. TITLE OF THE INVENTION**

**A SYSTEM FOR SWARM OF DRONE FOR CROWED BEHAVIOUR ANALYSIS**

**2. APPLICANT (S)**

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**3. PREAMBLE TO THE DESCRIPTION**

**COMPLETE SPECIFICATION**

The following specification particularly describes the invention and the manner in which it is to be performed.

## TECHNICAL FIELD

**[0001]** The present disclosure relates generally to the technical field of swarm of drone. In particular, it pertains to a system for aerial mapping for crowd behaviour and safety monitoring, and disaster response with minimal human intervention using swarm of unmanned aerial vehicles.

## BACKGROUND

**[0002]** Background description includes information that may be useful in understanding the present disclosure. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed disclosure, or that any publication specifically or implicitly referenced is prior art.

**[0003]** Conventional crowd behaviour analysis is generally based on visual inputs obtained from static or fixed location cameras including man-made surveillance. These processes are unable to perform persistent and continuous tracking thereby missing the continuous link. In the recent years, the use of unmanned aerial vehicles (UAVs) is increased in crowd monitoring and analysis for preventing stampede by estimating density and movement behaviour. Also, any suspicious activity among crowd can also be analysed for preventing incidents.

**[0004]** However, despite these advancements, the technologies continue to face several limitations such as the UAVs fleet are limited by fragmented functionalities such as some perform mapping assisted by SLAM/LiDAR technology while others perform surveillance. These frameworks are either centralized and face high latency, single point of failure or decentralized where no global optimization is possible. Existing wireless communication between peer-to-peer and to controlling are generally half-duplex and affected by severe degradation in disaster zones. Even high-bandwidth links for example, mmWave become bandwidth-constrained relative to the massive data volume of raw point-cloud streaming, necessitating architectural compression.

**[0005]** Recent advancements in the field have focused on improving by addressing above issues. One such Patent Document US20200394384A1

discloses a real-time aerial suspicious analysis system for identifying individual engaged in suspicious activities in public areas. The aerial suspicious analysis uses a drone for constant capturing and recording images/videos, and/or can be activated to capture/record based on a specific schedule and/or event; a YOLO detector to detect the individuals; a ScatterNet Hybrid Deep Learning (SHDL) network for individual pose estimation, and then classification is performed of the estimated pose to identify the suspicious/violent individuals.

**[0006]** While cited reference discloses a real-time aerial suspicious analysis system for identifying individual engaged in suspicious activities in public areas, there is a possibility to provide better solution for crowd monitoring that can address the abovementioned problem.

**[0007]** There is, therefore, a need to provide a simple, adaptive, and real-time decision making system while maintaining autonomous mapping, crowd monitoring that addresses abovementioned drawbacks.

## OBJECTS OF THE DISCLOSURE

**[0008]** A general object of the present disclosure is to provide a system employing hybrid swarm architecture for enabling globally optimal, self-organizing aerial missions for mapping, crowd safety monitoring, and disaster response with minimal human intervention.

**[0009]** An object of the present disclosure is to provide a system that uses full-duplex mesh networking to reduce link failures ensuring uninterrupted high-bandwidth data and command flow.

**[0010]** Another object of the present disclosure is to provide a system with dual layer architecture for low latency response time for anomaly detection, task reallocation, and emergency switching.

**[0011]** Another object of the present disclosure is to provide a system having simplified field deployment that reduces integration complexity and allows rapid deployment for disaster response and crowd monitoring operations. .

**[0012]** Another object of the present disclosure is to provide a system where peer-to-peer links with nearby drones and direct bidirectional link to the

centralised decision module, enabling collective situational awareness and low-latency coordination.

## SUMMARY

**[0013]** Aspects of the present disclosure relate to the technical field of swarm of drone. In particular, it pertains to a system for aerial mapping for crowd behaviour and safety monitoring, and disaster response with minimal human intervention using swarm of unmanned aerial vehicles. More particular, the proposed system is fusion of full-duplex mesh networking with hybrid MARL-CDM architecture for simultaneous mapping, anomaly detection, and adaptive task reallocation.

**[0014]** In an aspect, the disclosure is about a system to analyze crowed behaviors using swarm of drones includes a network of sensors including a stereo camera, LiDAR, inertial measurement unit, and a GPS module for localization, to capture real-time images of target locations; a human machine interface in communication with the drones to navigate and control flight path; and an onboard controller executing simultaneous localization and mapping, including at least one database, one or more processors coupled with memory, wherein the memory stores one or more instructions executable by the one or more processors to receive commands from the human-machine interface to take-off and position the drone at one or more target locations; maintain peer-to-peer links with nearby drones within the swarm of drone network; receive and record data from sensors for processing real-time captured images for preliminary crowed flow vectors at the target locations; transmit processed data to centralized decision module and collaborative SLAM module fusion using communication transceiver; detection and validation of anomalies to confirm cored hazards and emergency response; and continue mapping, avoidance of peer-to-peer collision, and failover handling if centralized decision module link is lost.

**[0015]** In an aspect, the on board controller includes Edge AI processor such as Nvidia Jetson OrinNano, intel Myraid, or Qualicomm flight RB5 for high-precision localization and 3D spatial perception.

**[0016]** In an aspect, the human machine interface operated by an operator, to operate command input panel, 3D map visualization, presentation of alert over in-built dash board alert dashboard, and data logging and reporting.

**[0017]** In an aspect, the communication transceiver is a bi-directional, self-healing full-duplex mesh network allowing all drones and the centralized decision module to exchange sensors data, control commands, and current status simultaneously.

**[0018]** In an aspect, the centralized decision module acts as the central intelligence core to perform global coordination, computation, and optimization.

**[0019]** In an aspect, the system collaborative SLAM fusion module merge 3D point clouds from all drones for drift correction and maintaining a unified spatial reference frame.

**[0020]** In an aspect, the system employ YOLO v8 in combination with spatial temporal graph convolutional networks (ST-GCN) for detecting crowd patterns from aerial perspective.

**[0021]** In an aspect, an integrated multi-agent reinforcement learning algorithm is used to dynamically allocate tasks across drones and for coverage, collision avoidance, balancing energy efficiency and integrating global mission objectives with real-time local autonomy.

**[0022]** In an aspect, the system follow adaptive mission switching upon anomaly detection and triggers instant switching between mapping, monitoring, and emergency response tasks without human intervention.

**[0023]** In an aspect, the system generates a semantic event token whenever detects a hazard, wherein all drones in swarm receives the token to update their local physics simulation layer via semantic logic to adapt standard compression to replace high-bandwidth geometry streaming.

**[0024]** Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0025]** The accompanying drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

**[0026]** FIG. 1 illustrates an exemplary block diagram depicting system architecture of hybrid multi-drones prototype, in accordance with embodiments of the present disclosure.

**[0027]** FIG. 2 illustrates an exemplary mission flow diagram of the hybrid multi-drones system of FIG. 1, in accordance with embodiments of the present disclosure.

## DETAILED DESCRIPTION

**[0028]** The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such details as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

**[0029]** Embodiments explained herein relate to the technical field of swarm of drone. In particular, it pertains to a system for aerial mapping for crowd behaviour and safety monitoring, and disaster response with minimal human intervention using swarm of unmanned aerial vehicles.

**[0030]** In an embodiment, the proposed disclosure is about a system to analyze crowed behaviors using swarm of drones includes a network of sensors 104 including a stereo camera, LiDAR, inertial measurement unit, and a GPS module; a human machine interface; and an onboard controller executing simultaneous localization and mapping, including at least one database, one or more processors to receive commands from the human-machine interface to take-off and position

the drone at one or more target locations; maintain peer-to-peer links within the swarm of drone network; receive and record data from sensors for processing real-time images for preliminary crowed flow vectors; transmit processed data to centralized decision module and collaborative SLAM module fusion using communication transceiver; detection and validation of anomalies to confirm cored hazards and emergency response; and continue mapping, avoidance of peer-to-peer collision, and failover handling if centralized decision module link is lost.

**[0031]** In an embodiment, the on board controller includes Edge AI processor such as Nvidia Jetson OrinNano, intel Myraid, or Qualicomm flight RB5 for high-precision localization and 3D spatial perception.

**[0032]** In an embodiment, the human machine interface operated by an operator, to operate command input panel, 3D map visualization, presentation of alert over in-built dash board alert dashboard, and data logging and reporting.

**[0033]** In an embodiment, the communication transceiver is a bi-directional, self-healing full-duplex mesh network allowing all drones and the centralized decision module to exchange sensors data, control commands, and current status simultaneously.

**[0034]** In an embodiment, the centralized decision module acts as the central intelligence core to perform global coordination, computation, and optimization.

**[0035]** In an embodiment, the system collaborative SLAM fusion module merge 3D point clouds from all drones for drift correction and maintaining a unified spatial reference frame.

**[0036]** In an embodiment, the system employ YOLO v8 in combination with spatial temporal graph convolutional networks (ST-GCN) for detecting crowd patterns from aerial perspective.

**[0037]** In an embodiment, an integrated multi-agent reinforcement learning algorithm is used to dynamically allocate tasks across drones and for coverage, collision avoidance, balancing energy efficiency and integrating global mission objectives with real-time local autonomy.

**[0038]** In an embodiment, the system follow adaptive mission switching upon anomaly detection and triggers instant switching between mapping, monitoring, and emergency response tasks without human intervention.

**[0039]** In an embodiment, the system generates a semantic event token whenever detects a hazard, wherein all drones in swarm receives the token to update their local physics simulation layer via semantic logic to adapt standard compression to replace high-bandwidth geometry streaming.

**[0040]** Referring to FIG. 1 where an exemplary block diagram depicting system 100 architecture of hybrid multi-drones prototype 102 is shown.

**[0041]** In an embodiment, the system 100 to analyze crowed behaviors using swarm of drones 102 includes a network of sensors 104 including a stereo camera 104-A, LiDAR 104-B, inertial measurement unit 104-C, and a GPS module 104-D for localization, to capture real-time images of target locations.

**[0042]** In an embodiment, the hierarchical hybrid architecture of the proposed swarm system 100, the UAV nodes 102-1, 102-2, …., 102-N collectively referred as UAV nodes 102, where each UAV node 102 (the drones 102 and UAVs 102 having same meaning and functions are used interchangeably in the disclosure, herein) is shown as an autonomous agent equipped with sensor network 104, an Edge AI Processor 106 including at least one database, one or more processors coupled with memory. The memory stores one or more instructions executable by the one or more processors. The swarm size can very from 3 to 50 UAVs 102 depend upon the mission complexity and mesh network stability. The speed of UAVs 102 can be 1 to 15 meter per seconds at altitude from 10 to 120 meters. The mission duration can be 20 to 120 minutes in one sortie.

**[0043]** The edge AI processor 106 executing local SLAM and YOLOv8-based crowd or hazard detection, and a local controller 108 responsible for flight stability and collision avoidance.

**[0044]** In an embodiment, the edge AI processor 106 executes navigational actions immediately via Optimistic Execution, providing 'Zero Perceived Latency' to the local loop. Simultaneously, a safety clamp validates these actions against immediate physical constraints. The on board controller 108 includes Edge AI

processor 106 such as Nvidia Jetson OrinNano, intel Myraid, or Qualicomm flight RB5 for high-precision localization and 3D spatial perception.

**[0045]** In an embodiment, The UAVs 102 communicate via a full-duplex mesh network 120**,** enabling simultaneous uplink of sensor and telemetry data to the Centralized Decision Module (CDM) 110 and downlink of optimized task assignments. Full-duplex refers to either true in-band full-duplex or effective full-duplex operation achieved via spatial frequency, or time-domain separation, ensuring continuous bidirectional flow. The communication range depends on the line of sight and is generally less than 5 kilometres. The data packet size is 50 to 500 Bytes per tick.

**[0046]** In an embodiment, the communication transceiver is a bi-directional, self-healing full-duplex mesh network allowing all drones 102 and the centralized decision module 110 to exchange sensors data, control commands, and current status simultaneously. Each UAV 102 communicates with its neighbours to propagate local map fragments and cooperative collision-avoidance data. Adaptive routing and link quality estimation ensure minimal packet loss under dynamic swarm motion.

**[0047]** In an embodiment, the CDM 110 contains a collaborative SLAM module 112 for global 3D map fusion, a Multi-Agent Reinforcement Learning (MARL) Optimizer 114 for dynamic mission allocation, the Anomaly Detection Engine (ADE) 116 for crowd and hazard detection, and the Hybrid Task Reallocation Core (HTRC) 118 for adaptive task reassignment. The centralized decision module 110 acts as the central intelligence core to perform global coordination, computation, and optimization.

**[0048]** In an embodiment, the CDM 110 is typically deployed on a GPU-accelerated edge server or one of the Leader UAVs equipped with higher compute capacity to reduce latency. The CDM 110 acts as the central intelligence core and performs global coordination, computation, and optimization. The system 100 collaborative SLAM fusion module 112 merge 3D point clouds from all drones 102 for drift correction and maintaining a unified spatial reference frame. The module 112 integrates LiDAR 104-B and visual data from all UAVs 102 to

maintain a single, consistent global 3D map, updating occupancy grids and structural reconstructions in real time;

**[0049]** In an embodiment, an integrated multi-agent reinforcement learning algorithm 114 is used to dynamically allocate tasks across drones 102 and for coverage, collision avoidance, balancing energy efficiency and integrating global mission objectives with real-time local autonomy. It also determines globally optimal task distributions, area partitioning, and reallocation strategies to minimize redundancy and energy usage.

**[0050]** In an embodiment, a semantic event core processes 'Event Tokens' such as Class:Fire, Vector:X,Y received from UAVs 102. Instead of processing raw video streams which consumes high bandwidth, it propagates these tokens to the Physics Engine to deterministically update the global simulation state. The token size is about 100Bytes.

**[0051]** In an embodiment, thehybrid task reallocation core (HTRC) 118 dynamically reassigns UAVs 102 to high-priority zones such as disaster points or crowd surges upon detection of anomalies, updating mission parameters on the fly.

**[0052]** In an embodiment, the processor employs CSLAM 112 that merges 3D point clouds from all UAVs 102, correcting for drift and maintaining a unified spatial reference frame. The YOLOv8 combined with ST-GCN used for detecting hazardous crowd patterns or collapsed structures from aerial perspective. A network health monitor continuously evaluates signal quality, latency, and throughput across the full-duplex mesh, rerouting data as needed. Further, a data synchronization manager timestamps and synchronizes all UAV sensor 104 inputs to maintain temporal consistency in the global map.

**[0053]** In an embodiment, the human machine interface 150 in communication with the drones 102 is operated by an operator to navigate and control flight paths of drones 102. The human machine interface 150 bridges human supervision with autonomous control, ensuring compliance with operational safety regulations

**[0054]** In an embodiment, the UAVs 102, mesh network 104, CDM 110, and HMI 150 are interconnected to form a cohesive, self-optimizing swarm system

100**.** HMI 150 provides 3D map visualization by real-time rendering of the global map generated by CSLAM 112, overlaid with hazard or density heat maps. It displays anomaly alerts, UAV health status, and mission progress in real time, and allows human override, mission reconfiguration, and manual task initiation when necessary. The HMI 150 archives flight telemetry and analytics for post-mission evaluation.

**[0055]** In an embodiment, the human machine interface 150 operated by an operator, to operate command input panel 156, 3D map visualization 154, presentation of alert over in-built dash board alert dashboard 152, and data logging and reporting.

**[0056]** FIG. 2 illustrates an exemplary mission flow diagram 200 of the hybrid multi-drones system 100. The flow diagram 200 represents the step-wise operational flow of the UAV swarm 102 during a mission. It begins, as per step 202, with UAV takeoff, self-diagnostics, sensor calibration, and mesh formation.

**[0057]** In an embodiment, each UAV 102 then performs local SLAM, edge processing, and crowd hazard detection as per step 204. According to step 206**,** transmitting processed data to the CDM 110 for fusion with CSLAM data.

**[0058]** In an embodiment, as per step 208, the CDM 110 executes CSLAM fusion to generate a global 3D map, and the MARL optimizer assigns UAVs to frontier points.

**[0059]** In an embodiment, in case the anomaly detected as per step 210 when a hazard or abnormal crowd movement is detected, as per step 212 and 214, the HTRC dynamically reallocates UAVs 102 to the anomaly**,** overriding non-critical tasks.

**[0060]** In an embodiment, if no anomaly is detected, then as per step 216, UAVs continue mapping autonomously. The flow captures both routine exploration and emergency response operations in a single coordinated flow.

**[0061]** In an embodiment, as per step 218, upon mission completion or safe crowd dispersion, CDM signals UAV return to base. The UAVs upload logs and HMI generates post-mission analytics including 3D terrain map, detected hazard zones, UAV flight paths, Communication statistics, MARL reward curves. This

enables simultaneous mapping, monitoring, and emergency response for achieving real-time, self-optimizing coordination with human-supervised transparency.

**[0062]** Thus, the disclosed hybrid multi-UAV system 100 operates through a set of iterative, multi-layered control loops, enabling autonomous mapping, crowd monitoring, and dynamic disaster response. Each loop is tightly integrated with the UAV nodes 102, the full-duplex mesh network 120, the Centralized Decision Module (CDM) 110, and the Human Machine Interface (HMI) 150. The system 100 ensures adaptive, real-time decision-making while maintaining local autonomy and global coordination.

**[0063]** While the foregoing describes various embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the invention is determined by the claims that follow. The invention is not limited to the described embodiments, versions or examples, which are included to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art.

## ADVANTAGES OF THE DISCLOSURE

**[0064]** The present disclosure provides a system employ a hybrid swarm architecture for enabling globally optimal, self-organizing aerial missions for mapping, crowd safety monitoring, and disaster response with minimal human intervention.

**[0065]** The present disclosure provides a system that uses full-duplex mesh networking to reduce link failures ensuring uninterrupted high-bandwidth data and command flow.

**[0066]** The present disclosure provides a system with dual layer architecture for low latency response time for anomaly detection, task reallocation, and emergency switching.

**[0067]** The present disclosure provides a system having simplified field deployment that reduces integration complexity and allows rapid deployment for disaster response and crowd monitoring operations. .

**[0068]** The present disclosure provides a system where peer-to-peer links with nearby drones and direct bidirectional link to the centralised decision module, enabling collective situational awareness and low-latency coordination.

**We Claim:**

1. A system (100) to analyze crowed behaviors using swarm of drones (102), the system (100) comprising:

a network of sensors (104) comprising a stereo camera (104-A), LiDAR (104-B), inertial measurement unit (104-C), and a GPS module (104-D) for localization, to capture real-time images of target locations;

a human machine interface (150) in communication with the drones (102) to navigate and control flight path; and

an onboard controller (108) executing simultaneous localization and mapping, comprising at least one database, one or more processors coupled with memory, wherein the memory stores one or more instructions executable by the one or more processors to:

receive commands from the human-machine interface (150) to take-off and position the drone (102) at one or more target locations;

maintain peer-to-peer links with nearby drones (102) within the swarm of drone network;

receive and record data from sensors (104) for processing real-time captured images for preliminary crowed flow vectors at the target locations;

transmit processed data to centralized decision module (110) and collaborative SLAM module fusion (112) using communication transceiver (120);

detection and validation of anomalies to confirm cored hazards and emergency response; and

continue mapping, avoidance of peer-to-peer collision, and failover handling if centralized decision module (110) link is lost.

2. The system as claimed in claim 1, wherein the on board controller (108) comprises Edge AI processor (106) such as Nvidia Jetson OrinNano, intel Myraid, or Qualicomm flight RB5 for high-precision localization and 3D spatial perception.

3. The system as claimed in claim 1, wherein the human machine interface (150), operated by an operator to operate command input panel (156), 3D map visualization (154), presentation of alert over in-built dash board alert dashboard (152), and data logging and reporting.

4. The system as claimed in claim 1, wherein the communication transceiver (120) is a bi-directional, self-healing full-duplex mesh network allowing all drones (102) and the centralized decision module (110) to exchange sensors data, control commands, and current status simultaneously.

5. The system as claimed in claim 1, wherein the centralized decision module (110) acts as the central intelligence core to perform global coordination, computation, and optimization.

6. The system as claimed in claim 1, wherein the system (100) collaborative SLAM fusion module (112) merge 3D point clouds from all drones (102) for drift correction and maintaining a unified spatial reference frame.

7. The system as claimed in claim 1, wherein the system (100) employ YOLO v8 in combination with spatial temporal graph convolutional networks (ST-GCN) for detecting crowd patterns from aerial perspective.

8. The system as claimed in claim 1, wherein an integrated multi-agent reinforcement learning algorithm (114) is used to dynamically allocate tasks across drones (102) and for coverage, collision avoidance, balancing energy efficiency and integrating global mission objectives with real-time local autonomy.

9. The system as claimed in claim 1, wherein the system (100) follow adaptive mission switching upon anomaly detection and triggers instant switching between mapping, monitoring, and emergency response tasks without human intervention.

10. The system as claimed in claim 1, wherein the system (100) generates a semantic event token whenever detects a hazard, wherein all drones (102) in swarm receives the token to update their local physics simulation layer via semantic logic to adapt standard compression to replace high-bandwidth geometry streaming.

**FOR VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI**



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**Dated: 12th May, 2026**

## ABSTRACT

## A SYSTEM FOR SWARM OF DRONE FOR CROWED BEHAVIOUR ANALYSIS

A system 100 to analyze crowed behaviors using swarm of drones 102 includes a network of sensors 104 including a stereo camera 104-A, LiDAR 104-B, inertial measurement unit 104-C, and a GPS module 104-D; a human machine interface 150; and an onboard controller 108 executing simultaneous localization and mapping, including at least one database, one or more processors to receive commands from the human-machine interface 150 to take-off and position the drone 102 at one or more target locations; maintain peer-to-peer links within the swarm of drone network; receive and record data from sensors 104 for processing real-time images for preliminary crowed flow vectors; transmit processed data to centralized decision module 110 and collaborative SLAM module fusion 112 using communication transceiver 120; detection and validation of anomalies to confirm cored hazards and emergency response; and continue mapping, avoidance of peer-to-peer collision, and failover handling if centralized decision module 110 link is lost.

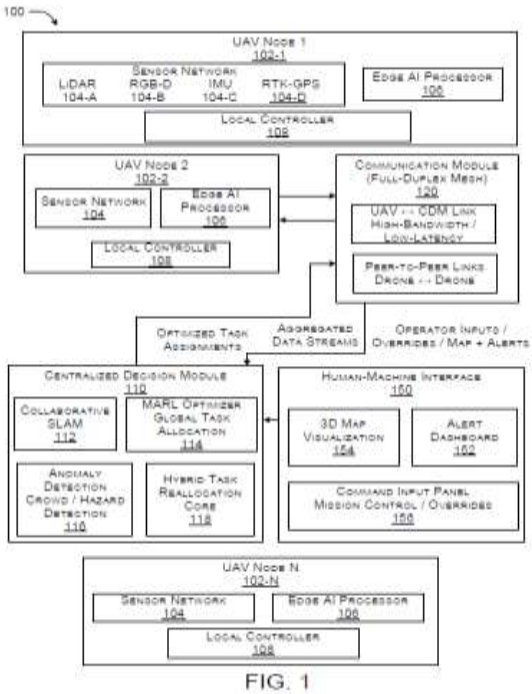


FIG. 1



Digitally Signed By:
TARUN KHURANA
Date: 12-05-2026 17:14:56